Hot Days Turn Deadlier Where Guns Are Easy to Carry

Lessons from “Access to Guns in the Heat of the Moment: More Restrictive Gun Laws Mitigate the Effect of Temperature on Violence” by Jonathan Colmer & Jennifer L. Doleac in *The Review of Economics and Statistics*, 2026.

Hot days lead to more gun homicides, but stricter concealed-carry laws sharply cut that effect. In today’s looser-law states, a year of days running one degree hotter than normal is estimated to cause roughly 299 more homicides than those same states would see under stricter laws.

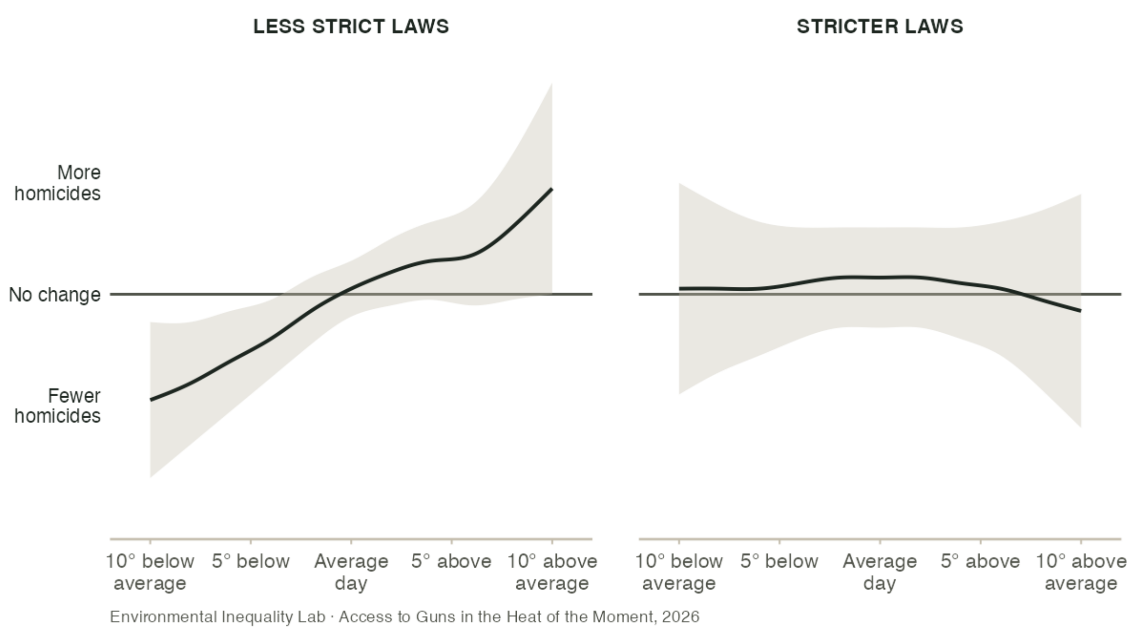
BACKGROUND

Gun violence is a large problem in the United States. Given the research that has established that high temperatures lead to increased aggression and violent behavior, this problem will only increase as temperatures rise due to climate change. Additionally, states differ significantly in their **concealed-carry laws** — the rules governing who can carry a hidden handgun in public. Some states let nearly any adult carry one; others require a permit that officials can deny, or ban concealed carry altogether. Together, these two facts raise an important question: **if heat makes people more likely to become violent, does having a gun close at hand change the results of these confrontations?** If so, then stricter concealed-carry laws may protect against an increase in homicides as temperatures rise.

To answer this question, researchers matched 25 years of daily police crime reports (1991–2016) with local weather records and each state’s concealed-carry laws, month by month. This allowed them to understand the temperature and laws in place at the time of every gun violence homicide.

HOW MUCH HOTTER DAYS RAISE HOMICIDES, DEPENDING ON HOW STRICT A STATE’S GUN LAWS ARE

Source: Colmer & Doleac (2026).



Note: Each panel shows the estimated relationship between daily temperature and homicides, after accounting for typical seasonal and local patterns. In states with less strict concealed-carry laws (left), hotter days correlate with more homicides. In states with stricter laws (right), the flatter line indicates that homicides do not increase with hotter weather. The shaded areas show the range of uncertainty around each estimate.

WHAT THE STUDY WAS

It is difficult to measure how a law might affect heat-induced gun violence because it is impossible to measure how a place might be different if a law had never been passed. The reason for a law might be directly related to a past event or a change in culture that could also change the amount of gun violence. It would also be difficult to compare states with different laws and/or temperatures for the same reason: there might be other circumstances (e.g. cultural, political, poverty) that would lead to different patterns in violence.

However, researchers found a solution that compared the amount of *extra* violence a hot day brought in strict-law places versus loose-law places compared to each state's own averages — shown in the figure on the previous page. Because the weather is often random, the researchers chose to make the assumption that despite other differences between states, there isn't a change in how people respond specifically to hot weather. Under that assumption, their comparison isolates how much a state's gun laws shape hot weather's effect on violence. As a check, they also tracked the same states over time, comparing each state's own hot-day homicides before and after it changed its own law. Both approaches point to the same conclusion.

FINDINGS

Through this design, researchers were able to conclude that the mitigating effect of gun homicides traces back to the law itself, not some other difference between the states being compared.

Putting numbers to that comparison: each one-degree-Celsius increase in daily temperature is tied to roughly five percent more homicides than usual on an average day in states with less strict concealed-carry laws. In states with stricter laws, that heat-driven increase is much smaller — both when comparing different states at the same time, and when tracking the same states before and after they changed their own laws. Nationally, the difference adds up: in today's looser-law states, a year of days running one degree hotter than normal is estimated to cause roughly 299 more homicides than those same states would see under stricter laws. **Stricter concealed-carry laws blunt most of the effect of heat on homicides.**

To put a number on these stakes, researchers used the standard estimates of the cost of a homicide to society (about $11.3 million each) to calculate that avoiding these heat-driven deaths nationwide would be worth roughly **$3.4 billion for every one-degree-Celsius rise in temperature.** This is comparable to the estimated social benefit of hiring nearly 13,000 more police officers. **While this is a rough estimation, the stakes of heat-driven gun violence are large.**

FOR MORE INFORMATION

Colmer, Jonathan, and Jennifer L. Doleac. 2026. "Access to Guns in the Heat of the Moment: More Restrictive Gun Laws Mitigate the Effect of Temperature on Violence." *Review of Economics and Statistics* 108 (1): 30–43. doi.org/10.1162/rest_a_01395.

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